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PSAC Exam Practice · Predicted Specimen Paper

2026

SCIENCE

Grade 5 · Paper 3

Animal Classification | Dissolving and Filtering | Forces

Time Allowed: **1 hour**

Total Marks: **50**

Name: _____

Class: _____ Date: _____

INSTRUCTIONS TO CANDIDATES

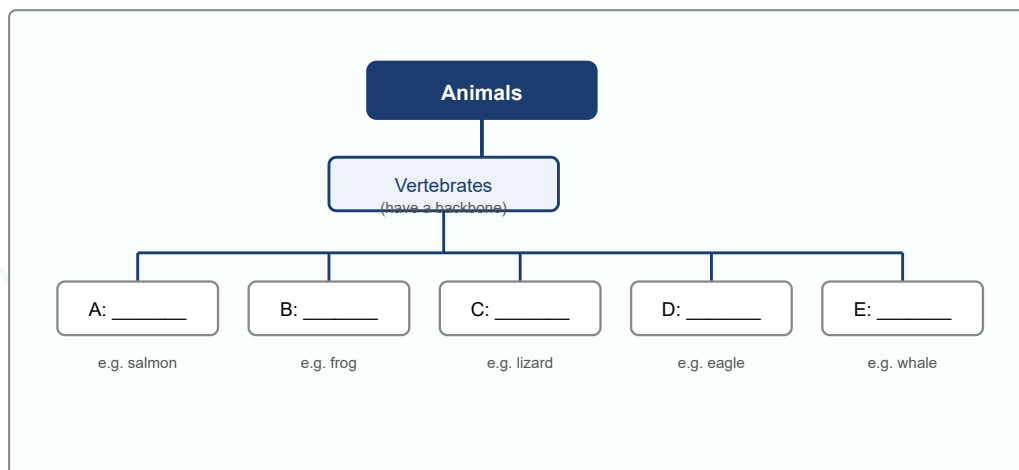
1. Write your name, class and date in the spaces provided above.
2. Answer **ALL** questions in **all three sections**.
3. Write your answers clearly in the spaces provided.
4. The mark for each question or part question is shown in square brackets [].
5. Diagrams should be clear and neatly labelled.

Section	Topic	Available	Score
A	Living Things — Animal Classification	20	
B	Matter and Materials — Dissolving and Filtering	15	
C	Energy and Forces — Push and Pull Forces	15	
TOTAL		50	

SECTION A — Living Things: Animal Classification [20 marks]

- 1 Study the animal classification diagram below. Use the words in the box to complete the labels A–E. [5]

amphibians birds fish mammals reptiles



- 2 Sort these animals by writing each one in the correct column of the table. [6]

butterfly dolphin eagle earthworm frog salmon

Vertebrate (has a backbone)	Invertebrate (no backbone)

[1 mark per animal correctly placed]

- 3 Tick **ONE** box to show the correct answer. [4]

(a) Which characteristic do ALL mammals share?

☐ they lay eggs ☐ they have feathers

☐ they feed their young on milk ☐ they live in water

(b) A frog is classified as an amphibian because it:

☐ has scales ☐ can live on both land and in water

☐ is warm-blooded ☐ has feathers

(c) Which of the following animals is an invertebrate?

☐ ☐ ☐ ☐

☐ shark ☐ snake ☐ spider ☐ sparrow

(d) Birds are warm-blooded animals. This means their body temperature:

☐ changes with the environment

☐ stays constant regardless of the environment

☐ is always lower than 0°C ☐ is always exactly 100°C

4 Write **TRUE** or **FALSE** for each statement about animal classification.

[3]

Statement	True or False
All reptiles are cold-blooded and have dry, scaly skin.	
A whale is classified as a fish because it lives in the sea.	
Insects have six legs and three body parts.	

5 A student finds an animal with the following features: *it has 8 legs, no wings, no backbone, and a body divided into two parts.*

(a) To which group of invertebrates does this animal belong?

[1]

(b) Give **one** example of an animal from this group.

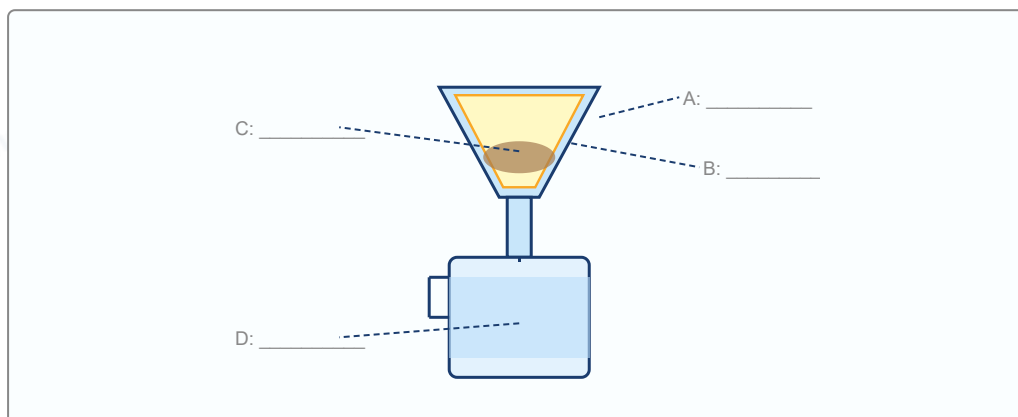
[1]

SECTION B — Matter and Materials: Dissolving and Filtering [15 marks]

- 6 The diagram below shows a filtration apparatus. Label the parts A–D.

filter funnel filter paper filtrate residue

[4]



- 7 A student adds different substances to water and stirs. She records whether each substance dissolves.

[4]

Substance	Dissolves in water?
Salt	Yes
Sand	No
Sugar	Yes
Chalk powder	No

- (a) What is the correct scientific word for a substance that dissolves in water?

[1]

- (b) A mixture of salt and sand is passed through filter paper. Which substance would pass through the filter paper? Give a reason.

[2]

- (c) After filtering the salt solution, how could you get the salt back?

[1]

8 Tick **ONE** box to show the correct answer.

[4]

(a) When sugar dissolves in water, it forms a:

☐ suspension ☐ solution ☐ compound ☐ element

(b) Which method would BEST separate a mixture of sand and water?

☐ evaporation ☐ filtration ☐ melting ☐ dissolving

(c) Heating salt water until the water evaporates is used to recover:

☐ the sand ☐ the salt ☐ the water ☐ the air

(d) Which of the following will dissolve fastest in hot water?

☐ large sugar crystals in cold water

☐ crushed sugar (powder) in hot water

☐ large sugar crystals without stirring ☐ sand in hot water

9 Explain why a teaspoon of sugar dissolves faster in hot water than in cold water.

[3]

SECTION C — Energy and Forces: Push and Pull Forces [15 marks]

- 10 Look at the diagram below showing four everyday situations. For each situation, state whether the force applied is a **push**, a **pull**, or **both**. [4]

Situation	Push / Pull / Both
Kicking a ball	
Opening a drawer	
Squeezing toothpaste from a tube	
Rowing a boat with oars	

- 11 Answer these questions about forces. [4]

(a) What is the name of the force that pulls all objects towards the Earth? [1]

(b) Name the force that slows down a moving object when two surfaces rub together. [1]

(c) What is the unit used to measure force? [1]

(d) Name the instrument used to measure force. [1]

- 12 Tick **TRUE** or **FALSE** for each statement about forces. [4]

Statement	True	False
A force can change the speed, direction or shape of an object.	☐	☐
Gravity only acts on very heavy objects.	☐	☐
Friction always acts in the direction opposite to motion.	☐	☐
A smooth surface produces more friction than a rough surface.	☐	☐

- 13** A student pushes a heavy box across a wooden floor. She finds it easier to push when the floor is very smooth.

(a) Explain why the box is easier to push on a smooth floor.

[2]

(b) Suggest **one** way to increase the friction between the box and the floor.

[1]

END OF PAPER 3 · Total Marks: 50

MARK SCHEME — Grade 5 Science · Paper 3

For teacher / parent use only. Accept any reasonable scientific answer that conveys the same meaning. Award full marks where a candidate's response is scientifically correct even if worded differently. Do not penalise poor spelling unless it changes the scientific meaning.

SECTION A — Animal Classification [20 marks]

Q	Part	Acceptable Answer	Marks
1	A–E	A: fish · B: amphibians · C: reptiles · D: birds · E: mammals (1 mark each)	5
2	—	Vertebrates: dolphin, eagle, frog, salmon · Invertebrates: butterfly, earthworm (1 mark each correctly placed)	6
3	(a)	they feed their young on milk	1
	(b)	can live on both land and in water	1
	(c)	spider	1
	(d)	stays constant regardless of the environment	1
4	—	TRUE / FALSE / TRUE (1 mark each)	3
5	(a)	Arachnids	1
	(b)	Any correct arachnid: spider / scorpion / tick / mite	1

SECTION B — Dissolving and Filtering [15 marks]

Q	Part	Acceptable Answer	Marks
6	A–D	A: filter funnel · B: filter paper · C: residue · D: filtrate (1 mark each)	4
7	(a)	soluble	1
	(b)	Salt — because salt is soluble (it dissolved) so it passes through the filter paper with the water. Sand stays on the filter paper as it is insoluble. [1 mark for salt, 1 mark for reason]	2
	(c)	Evaporate / heat the water (until all the water evaporates and the salt remains).	1
8	(a)	solution	1
	(b)	filtration	1
	(c)	the salt	1
	(d)	crushed sugar (powder) in hot water	1
9	—	Hot water has more energy / the water particles move faster [1]. This causes the sugar particles to break apart and spread through the water more quickly [1]. Any 2 correct scientific points earn 2 marks; 3rd mark for a clear, logical explanation. Accept: "particles in hot water	3

Q	Part	Acceptable Answer	Marks
		move faster so they collide with the sugar more often, breaking it down faster."	

SECTION C — Forces [15 marks]

Q	Part	Acceptable Answer	Marks
10	—	Kicking a ball: push · Opening a drawer: pull · Squeezing toothpaste: push · Rowing a boat: both (accept pull for rowing) (1 mark each)	4
11	(a)	Gravity	1
	(b)	Friction	1
	(c)	Newton (N)	1
	(d)	Force meter / Newton meter / spring balance	1
12	—	TRUE / FALSE / TRUE / FALSE (1 mark each)	4
13	(a)	A smooth floor has less friction (less resistance) between the box and the floor, so less force is needed to push it. [1 mark for "less friction", 1 mark for "less force needed"]	2
	(b)	Any correct answer: put a rough mat under the box / add sandpaper to the floor / wet the surface (accept any reasonable method that increases roughness)	1